

2023

Time : 3 hours

Full Marks : 70

*Candidates are required to give their answers
in their own words as far as practicable.*

The figures in the margin indicate full marks.

Answer all sections as directed.

Section-A

Objective type Questions

Compulsory

1. Choose the correct alternative for each of the
following questions : $2 \times 10 = 20$

(a) If $S=R=1$, in SR flip-flop then the next
state is :

Predictable

☒ Unpredictable

Complement of previous state

None of these

- (b) Which of the following is not a primary function of the CPU ?

Fetch

Decode

Execute

☒ Output

- (c) Which of the following is/are in POS form :

☒ $(X' + Z)(W' + Z')$

$X'Z + W'Z'$

Both of these

None of these

- (d) Which addressing mode uses a constant value within the instruction to directly specify the operand ?

☒ Immediate

Indirect

Register

Direct

- (e) Which of the following circuit is used to store one bit of data ?

☒ Flip flop

Decoder

Encoder

Register

- (f) Subtraction in computer is carried out by :

1's complement

☒ 2's complement

9's complement

None of these

- (g) In which of the following form the computer stores its data in memory ?

Hexadecimal form

Odal form

☒ Binary form

Decimal form

- (h) Which of the following is a combinational circuit which sends data from a single source to two or more separate destination :

Multiplexer

☒ Demultiplexer

Encoder

Decoder

- (i) Don't care condition is denoted by X in k-map. What is the value for X ?

0 and 1

☒ Either 0 or 1

1 and 2

None of these

- (j) Which register holds the memory address of the instruction that is currently being executed by the CPU ?

☒ Memory address register

Program counter

Memory data register

Insruction register

Section-B

Short Answer type Questions

2. Answer any four questions of the following :

5×4=20

(a) Draw the circuit diagram for the following expressions :

(i) $(a+b).(a'+b).(a'+b')$

(ii) $AB+B'C+A'C'$

(b) How to obtain NOR gate using NAND gate ? Define universal gate.

(c) What is the main disadvantage of SR flip flop ? How to resolve this disadvantages ?
Write the characteristic table of SR flip flop.

(d) Prove the following expressions by algebraic method :

(i) $A'B(D'+CD)+B(A+A'CD)=B$

(ii) $(xy'+x'y)'=xy+x'y'$

(e) Explain half adder with truth table and suitable circuit diagram.

(f) Define the following terminology :

(i) Maxterm

(ii) Sequential circuit ✓

(iii) Full adder ✓

(iv) Encoder ✓

(v) K-map ✓

Section-C

Long Answer type Questions

3. Answer any two questions of the following :

15×2=30

(a) Define multiplexer. Design 8 to 1 multiplexer with suitable truth table.

(b) Convert the following into product of sum form :

$$X'Y+Y(Z'(Z'+Y))$$

(c) Simplify the following boolean function F together with the don't care condition d in

(i) sum of product form

(ii) product of sum form

$$F(w, x, y, z) = \sum(1, 3, 7, 11, 15)$$

$$d(w, x, y, z) = \sum(0, 2, 5) \quad (x' + y)(w' + z)$$

(d) Define flip flop. Write most common application of flip flop. Explain the master slave flip flop.

_____ x _____